

Arnold tongue of two coupled PING circuits

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Abstract

This experiment will attempt to reproduce a subset of Lowet et al. [1]: the Arnold tongue formed by two reciprocally coupled PING circuits as their natural-frequency detuning and coupling strength are varied. An Arnold tongue is the region in which coupled oscillators adopt a common frequency despite having different uncoupled natural frequencies. The result has two control axes—uncoupled natural-frequency detuning Δf_0 and coupling strength K —and one measured response axis, the fraction of valid trials classified as frequency- and phase-locked. The reproduction is restricted to the paper’s two-circuit synchronization result. It does not test the spatially extended network, natural-image reconstruction, or proposed phase and frequency codes. Two independently driven PING circuits will be simulated across the two control axes. A successful reproduction will recover a contiguous locked region centred near zero detuning that widens as coupling increases, with the intrinsically faster circuit leading in phase inside that region.

Methods

1. **Construct two independently driven PING circuits.** Circuits A and B will each contain 80 conductance-based leaky integrate-and-fire excitatory neurons and 20 inhibitory neurons. Within each circuit, excitatory neurons will project to inhibitory neurons through AMPA synapses, and inhibitory neurons will project to excitatory neurons through GABA-A synapses. The circuits will use identical neuron, synapse, connectivity, and within-circuit weight parameters.

Independent Poisson spike populations will drive the two excitatory populations. Input spikes will be independently sampled between circuits and trials; no input spikes will be shared. The network parameter seed will remain fixed across the primary sweep so that detuning and coupling are not confounded with different weight realizations.

Reciprocal coupling will match the two-circuit configuration of Lowet et al. [1]:

$$E_A \rightarrow E_B, \quad E_A \rightarrow I_B, \quad E_B \rightarrow E_A, \quad E_B \rightarrow I_A.$$

All four cross-circuit projections will use AMPA synapses. Cross-circuit E-to-E and E-to-I weights will be controlled by the same coupling parameter K and varied together. The uncoupled graph will retain all four projections with zero-valued weights, preserving graph topology across conditions.

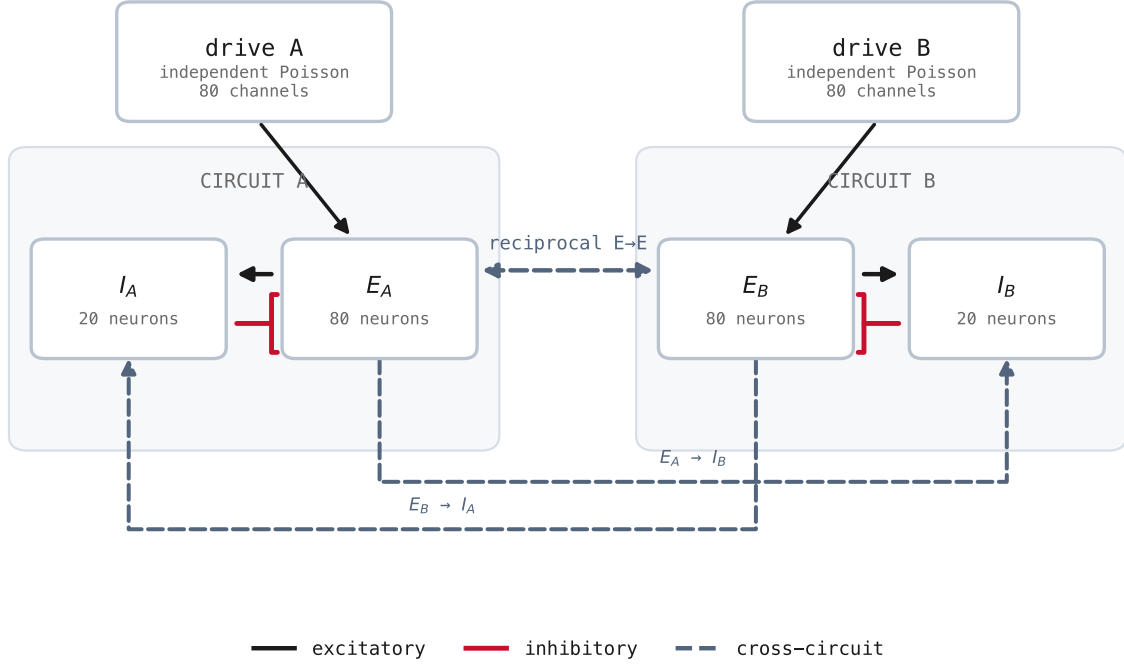


Figure 1: Network graph for the reproduction. Each independent input drives one local E $\leftrightarrow$ I PING loop. Dashed projections are the four reciprocal cross-circuit AMPA pathways varied together by K . Red bar-headed projections are local GABA-A inhibition.

2. **Calibrate drive against uncoupled gamma frequency.** One uncoupled PING circuit will be simulated across a bounded input-rate grid. The operating range will be the contiguous valid interval in which gamma peak frequency increases monotonically with input rate. The calibration will be completed before coupled conditions are inspected. Pairs of input rates will be selected to sample signed target detunings of approximately

$$\Delta f_0 \in \{-6, -4, -3, -2, -1, -0.5, 0, 0.5, 1, 2, 3, 4, 6\} \text{ Hz.}$$

For every input pair and trial seed, the two circuits will first run with $K = 0$. Their measured post-transient frequencies will define natural- frequency detuning:

$$\Delta f_0 = f_A^0 - f_B^0.$$

This measurement, rather than input-rate difference or coupled frequency difference, will determine the horizontal coordinate of every coupled result.

3. **Register and execute the detuning-by-coupling sweep.** A bounded pilot at zero detuning and at the largest positive and negative target detunings will locate a coupling interval containing uncoupled, partially locked, and locked behaviour without silencing or saturating either circuit. The pilot will set only the limits of K and will not enter the primary analysis.

The primary grid will contain $K = 0$ and ten equally spaced nonzero values over the selected interval. Every measured detuning will be crossed with every coupling value and at least five independently generated pairs of input spike trains. No grid point will be added or removed after inspecting the primary locking map.

Simulations will use a 0.1 ms timestep and run for 3 s. The first 500 ms will be excluded from analysis. Population spikes, membrane voltages, and projection conductances will be retained with the compiled graph, generated inputs, parameter tensors, run configuration, and seed ledger.

A trial will be invalid if any recorded state is non-finite, either mean excitatory or inhibitory firing rate is below 1 Hz, or either excitatory population has no spectral peak between 25 and 80 Hz. The same validity rule will be applied to every sweep cell.

4. **Measure frequency and relative phase.** Excitatory population spikes will be converted to population rates by Gaussian smoothing with a 5 ms standard deviation. Gamma peak frequency will be estimated from the post-transient rate spectrum. Instantaneous phase will be calculated from the analytic signal after zero-phase band-pass filtering from 25 to 90 Hz. The filter will remain fixed across conditions.

Relative phase will be

$$\varphi(t) = \text{unwrap}(\theta_{A(t)} - \theta_{B(t)}).$$

Each trial will report the two emergent frequencies, their absolute difference, the linear slope of $\varphi(t)$, the number of complete 2π phase slips, phase-locking value, and circular mean phase difference.

5. **Classify locking without using phase-locking value alone.** Frequency, drift, and phase-slip tolerances will be fixed from estimator variability in repeated zero-detuning and uncoupled calibration trials before the primary sweep. A valid trial will be classified as locked only if its emergent frequency difference, absolute relative-phase slope, and phase-slip count all fall within their registered tolerances. Phase-locking value will describe locking strength but will not independently determine the classification.
6. **Reconstruct and test the Arnold tongue.** The primary result will be the fraction of valid trials classified as locked at each measured Δf_0 and K . Supporting maps will show emergent frequency difference, relative- phase slope, phase-slip rate, phase-locking value, and circular mean phase difference on the same grid.

The reproduction will pass if the locking map contains a contiguous region centred near zero detuning whose width increases across at least three successive nonzero coupling levels. Within locked nonzero-detuning conditions, the circuit with the greater uncoupled natural frequency must lead in phase. It will fail if locking is absent, confined to zero detuning, does not widen with coupling, or occurs only in invalid conditions.

Representative traces will be selected by fixed grid location: zero detuning, one nonzero-detuning condition inside the locking region, and the same detuning immediately outside it. Each trace will show both excitatory rasters, both excitatory population rates, unwrapped relative phase, and sliding-window frequency difference.

References

1. Lowet, Roberts, Hadjipapas, Peter, van der Eerden & De Weerd: *Input-Dependent Frequency Modulation of Cortical Gamma Oscillations Shapes Spatial Synchronization and Enables Phase Coding*. PLOS Computational Biology, 2015. doi:10.1371/journal.pcbi.1004072